Lab 6: Electronic Door Lock

Today, we will dive into finite state machines in VHDL. We will do so by implementing the simplex electronic door lock.

Discussion – Lab 5

1. Any issues?
2. Solutions to Lab 5

Task – Electronic Door Lock

This task will be part of journal 2.

1. Draw out the FSM diagram of a debouncer. Using the FSM template, write a *debouncer* entity, figure 1 in file *debouncer.vhd*
    a. Test and verify in simulation
    b. Bonus: Imagine that the input takes longer than one clock cycle to settle. Redraw the FSM diagram and implement your debouncer to encounter this. Hint: Use a shift register and make the width of the shift register generic.

Figure 1: Debouncer block diagram

Figure 2: Electronic door lock block diagram

2. Draw out the FSM diagram of a simplex electronic door lock. Using the FSM template, write a simplex electronic door lock controller *edl_controller* entity, figure 2, in file *edl_controller.vhd*
    a. Test and verify in simulation
3. In block design, implement the full electronic door lock system and your *keypad_driver* from lab 4
    a. Create a testbench
    b. Test and verify in simulation

*Note: A common mistake with state machines in .vhdl is '**Unresolved port "port_name" is multiply driven**.' This is caused when multiple processes have the ability to assign a value to a port, even if both processes logically cannot assign values at the same time. The solution is to only ever assign values to a specific port in one process.*